Class 12 Mathematics

Chapter 4: Determinants

School Worksheet: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to:

(A) 6 **(B)** ± 6 **(C)** -6 **(D)** 0

Formula / idea: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

1 Expand the left determinant.

$$\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = x^2 - 36$$

2 Expand the right determinant.

$$\begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix} = 36 - 36 = 0$$

3 Equate the two and solve. ✓ 1

$$x^2 - 36 = 0 \Rightarrow x^2 = 36 \Rightarrow x = \pm 6$$

4 Hence the correct option is (B).

Answer **(B)** ± 6

Watch out: Writing only $x = 6$; a square equation gives two roots, ± 6.

2 MCQ 1 mark

If A is a square matrix of order 3 with $|A| = 5$, then $|\text{adj } A|$ is:

(A) 5 **(B)** 25 **(C)** 125 **(D)** $\frac{1}{5}$

Formula / idea: For a square matrix of order n, $|\text{adj } A| = |A|^{n-1}$.

1 Use the property of the adjoint.
$|\text{adj } A| = |A|^{n-1}$

2 Put $n = 3$ and $|A| = 5$. ✓ 1
$|\text{adj } A| = 5^{3-1} = 5^2 = 25$

3 Hence the correct option is (B).

Answer **(B)** 25

Watch out: Using $|A|^n = 125$; the power is $n - 1$, not n.

3 MCQ 1 mark

If A is a square matrix of order 3 and $|A| = 4$, then $|2A|$ is:

(A) 8 **(B)** 16 **(C)** 32 **(D)** 64

Formula / idea: For a square matrix of order n, $|kA| = k^n|A|$.

1 Use the property for a scalar multiple.
$|kA| = k^n|A|$

2 Put $k = 2$, $n = 3$, $|A| = 4$. ✓ 1
$|2A| = 2^3 \times 4 = 8 \times 4 = 32$

3 Hence the correct option is (C).

Answer **(C)** 32

Watch out: Writing $|2A| = 2|A| = 8$; each of the 3 rows is multiplied by 2.

4 MCQ 1 mark

The area of the triangle with vertices $(-2,-3)$, $(3,2)$ and $(-1,-8)$ is:

(A) 15 **sq units** **(B)** 30 sq units

(C) 20 sq units **(D)** 10 sq units

Formula / idea: Area $\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$, taken with absolute value.

1 Write the area as a determinant.

$$\Delta = \frac{1}{2}\begin{vmatrix} -2 & -3 & 1 \\ 3 & 2 & 1 \\ -1 & -8 & 1 \end{vmatrix}$$

2 Expand along R_1.

$$\Delta = \tfrac{1}{2}\left[-2(2+8)+3(3+1)+1(-24+2)\right]$$
$$= \tfrac{1}{2}(-20+12-22) = \tfrac{1}{2}(-30) = -15$$

3 Area is always positive, so take the absolute value. ✓ 1

$$\text{Area} = |-15| = 15 \text{ sq units}$$

4 Hence the correct option is (A).

Answer **(A)** 15 **sq units**

Watch out: Forgetting the $\frac{1}{2}$ and choosing 30 sq units.

5 MCQ 1 mark

The cofactor of the element a_{12} in $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$ is:

(A) 46 **(B)** -46 **(C)** 4 **(D)** -4

Formula / idea: Cofactor $A_{ij} = (-1)^{i+j}M_{ij}$, where M_{ij} is the minor.

1 Delete row 1 and column 2 to get the minor M_{12}.

$$M_{12} = \begin{vmatrix} 6 & 4 \\ 1 & -7 \end{vmatrix} = -42 - 4 = -46$$

2 Attach the sign $(-1)^{1+2}$. ✓ 1

$$A_{12} = (-1)^{1+2}M_{12} = -(-46) = 46$$

3	Hence the correct option is (A).	

Answer **(A)** 46

Watch out: Giving the minor -46 as the answer and forgetting the sign $(-1)^{i+j}$.

6 **Assertion-Reason** 1 mark

Assertion (A): If A is a 3×3 matrix with $|A| = 2$, then $|A\,(\text{adj}\,A)| = 8$. **Reason (R):** For any square matrix A, $A\,(\text{adj}\,A) = |A|\,I$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $A(\text{adj}\,A) = (\text{adj}\,A)A = |A|\,I$ for every square matrix A.

1	Check A: use $A(\text{adj}\,A) = \|A\|I$ with $n = 3$. $\|A(\text{adj}\,A)\| = \big\|\,\|A|\,I\,\big\| = |A|^3|I| = 2^3 = 8$	
2	So Assertion (A) is true.	
3	Check R: $A(\text{adj}\,A) = \|A\|I$ is a standard theorem, so R is true.	
4	A is obtained directly from R, so R correctly explains A.	
5	Hence the correct option is (A).	✓ 1

Answer **(A) Both A and R are true, and R is the correct explanation of A**

Watch out: Taking $\big||A|I\big| = |A| = 2$; a scalar comes out of a 3×3 determinant as $|A|^3$.

7 **Short answer** 2 marks

Find the values of k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$ and $(0, 2)$ is 4 sq units.

Formula / idea: Area $= \pm\dfrac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$, so use both signs.

1	Write the area as a determinant. $\Delta = \dfrac{1}{2}\begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix}$	✓ ½

2 Expand along R_1. ✓ ½

$$\Delta = \frac{1}{2}\left[k(0-2) - 0 + 1(8-0)\right] = \frac{1}{2}(8-2k) = 4-k$$

3 Area is 4, so the determinant value can be $+4$ or -4. ✓ ½

$$4-k = \pm 4$$

4 Solve each case.

$$4-k=4 \Rightarrow k=0; \qquad 4-k=-4 \Rightarrow k=8$$

5 Hence $k=0$ or $k=8$. ✓ ½

Answer $k=0$ **or** $k=8$

Watch out: Taking only $4-k=4$ and missing the second value $k=8$.

Level 2

17 marks

8 **Short answer** 2 marks

Find the adjoint and inverse of $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$.

Formula / idea: $A^{-1} = \dfrac{1}{|A|}\operatorname{adj} A$, provided $|A| \neq 0$.

1 For a 2×2 matrix, swap the diagonal and change signs of the others.

$$\operatorname{adj}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

2 Write the adjoint of A. ✓ 1

$$\operatorname{adj} A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$$

3 Find $|A|$. ✓ ½

$$|A| = 2(3) - (-1)(4) = 6+4 = 10 \neq 0$$

$|A| \neq 0$, so A^{-1} exists.

4 Write the inverse. ✓ ½

$$A^{-1} = \frac{1}{|A|}\operatorname{adj} A = \frac{1}{10}\begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$$

Answer $\operatorname{adj} A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$, $A^{-1} = \dfrac{1}{10}\begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$

Watch out: Changing the signs of the diagonal elements instead of the off-diagonal ones.

9 **Short answer** 3 marks

If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1}.

Formula / idea: Multiply the matrix equation by A^{-1} and use $A^{-1}A = I$.

1 Find $A^2 = A \cdot A$. ✓ 1

$$A^2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 9-1 & 3+2 \\ -3-2 & -1+4 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

2 Write $5A$ and $7I$.

$$5A = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix}, \qquad 7I = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

3 Substitute in $A^2 - 5A + 7I$. ✓ 1

$$A^2 - 5A + 7I = \begin{bmatrix} 8-15+7 & 5-5+0 \\ -5+5+0 & 3-10+7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

4 Hence $A^2 - 5A + 7I = O$ is shown.

5 Check that A^{-1} exists.

$$|A| = 6 + 1 = 7 \neq 0$$

6 Pre-multiply $A^2 - 5A + 7I = O$ by A^{-1}. ✓ ½

$$(A^{-1}A)A - 5(A^{-1}A) + 7A^{-1}I = A^{-1}O \;\Rightarrow\; A - 5I + 7A^{-1} = O$$

7 Make A^{-1} the subject and simplify. ✓ ½

$$A^{-1} = \frac{1}{7}(5I - A) = \frac{1}{7}\begin{bmatrix} 5-3 & 0-1 \\ 0+1 & 5-2 \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

Answer $A^2 - 5A + 7I = O$ **shown;** $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

Watch out: Writing $5 - A$ instead of $5I - A$; a number cannot be subtracted from a matrix.

10 **Short answer** 2 marks

Using determinants, find the equation of the line joining the points $(1, 2)$ and $(3, 6)$.

Formula / idea: Three points are collinear when the area determinant is zero.

1 Let $P(x, y)$ be any point on the line.

2 P, $(1, 2)$, $(3, 6)$ are collinear, so the area is zero. ✓ 1

$$\frac{1}{2}\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 0$$

3 Expand along R_1.
$x(2-6) - y(1-3) + 1(6-6) = 0 \Rightarrow -4x + 2y = 0$ ✓ ½

4 Simplify to get the required line.
$y = 2x$ ✓ ½

Answer $y = 2x$

Watch out: Expanding the determinant with wrong signs; the middle term of R_1 is $-y(\ldots)$.

11 Short answer — 3 marks

Solve the system of equations using the matrix method: $5x + 2y = 4$, $7x + 3y = 5$.

Formula / idea: If $AX = B$ and $|A| \neq 0$, then $X = A^{-1}B$ (unique solution).

1 Write the system as $AX = B$.
$\begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ ✓ ½

2 Find $|A|$.
$|A| = 15 - 14 = 1 \neq 0$
So the system has a unique solution. ✓ ½

3 Find $\operatorname{adj} A$ and A^{-1}.
$\operatorname{adj} A = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}, \qquad A^{-1} = \frac{1}{1}\begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$ ✓ 1

4 Find $X = A^{-1}B$.
$X = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 - 10 \\ -28 + 25 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ ✓ ½

5 Hence $x = 2$, $y = -3$. ✓ ½

Answer $x = 2,\ y = -3$

Watch out: Writing $X = BA^{-1}$; the order must be $A^{-1}B$.

12 Short answer — 2 marks

Examine the consistency of the system: $x + 3y = 5$, $2x + 6y = 8$.

Formula / idea: If $|A| = 0$ and $(\operatorname{adj} A)B \neq O$, the system is inconsistent.

1 Write the system as $AX = B$.
$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$

2 Find $|A|$.
$|A| = 6 - 6 = 0$
A is singular, so check $(\text{adj}\, A)B$. ✓ ½

3 Find $\text{adj}\, A$.
$\text{adj}\, A = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix}$ ✓ ½

4 Find $(\text{adj}\, A)B$.
$(\text{adj}\, A)B = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 30 - 24 \\ -10 + 8 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \neq O$ ✓ ½

5 Hence the system is inconsistent (it has no solution). ✓ ½

Answer **Inconsistent (no solution), since $|A| = 0$ and $(\text{adj}\, A)B \neq O$**

Watch out: Stopping at $|A| = 0$; you must also check $(\text{adj}\, A)B$ before concluding.

13 **Short answer** 2 marks

If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, show that $A^{-1} = \frac{1}{19}A$.

Formula / idea: $A^{-1} = \frac{1}{|A|}\,\text{adj}\, A$

1 Find $|A|$.
$|A| = 2(-2) - 3(5) = -4 - 15 = -19 \neq 0$ ✓ ½

2 Find $\text{adj}\, A$.
$\text{adj}\, A = \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix}$ ✓ ½

3 Write A^{-1}.
$A^{-1} = \frac{1}{-19}\begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix}$ ✓ ½

4 Take the minus sign inside the matrix.
$A^{-1} = \frac{1}{19}\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} = \frac{1}{19}A$ ✓ ½

5 Hence $A^{-1} = \frac{1}{19}A$ is shown.

Answer $A^{-1} = \frac{1}{19}\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} = \frac{1}{19}A$ **(shown)**

Watch out: Losing the negative sign of $|A| = -19$ when writing A^{-1}.

14 **Short answer** 3 marks

If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.

Formula / idea: Reversal law: $(AB)^{-1} = B^{-1}A^{-1}$.

1 Find AB. ✓ ½

$$AB = \begin{bmatrix} 2-3 & -4+9 \\ 1+4 & -2-12 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 5 & -14 \end{bmatrix}$$

2 Find $|AB|$ and $\operatorname{adj}(AB)$. ✓ ½

$$|AB| = 14 - 25 = -11, \qquad \operatorname{adj}(AB) = \begin{bmatrix} -14 & -5 \\ -5 & -1 \end{bmatrix}$$

3 Write $(AB)^{-1}$. ✓ ½

$$(AB)^{-1} = \frac{1}{-11}\begin{bmatrix} -14 & -5 \\ -5 & -1 \end{bmatrix} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$$

4 Find A^{-1}. ✓ ½

$$|A| = -8 - 3 = -11, \qquad A^{-1} = \frac{1}{-11}\begin{bmatrix} -4 & -3 \\ -1 & 2 \end{bmatrix} = \frac{1}{11}\begin{bmatrix} 4 & 3 \\ 1 & -2 \end{bmatrix}$$

5 Find B^{-1}. ✓ ½

$$|B| = 3 - 2 = 1, \qquad B^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

6 Multiply $B^{-1}A^{-1}$. ✓ ½

$$B^{-1}A^{-1} = \frac{1}{11}\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 1 & -2 \end{bmatrix} = \frac{1}{11}\begin{bmatrix} 12+2 & 9-4 \\ 4+1 & 3-2 \end{bmatrix} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$$

7 Both sides are equal, so $(AB)^{-1} = B^{-1}A^{-1}$ is verified.

Answer $(AB)^{-1} = B^{-1}A^{-1} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$ **(verified)**

Watch out: Multiplying $A^{-1}B^{-1}$ instead of $B^{-1}A^{-1}$; matrix order matters.

Level 3

29 marks

15 **Long answer** 5 marks

A school wants to allocate students into three clubs: Sports, Music and Drama, under the following conditions: (i) the number of students in the Sports club should be equal to the sum of the numbers of students in the Music and Drama clubs; (ii) the number of students in the Music club should be 20 more than half the number of students in the Sports club; (iii) the total number of students allocated to the three clubs is 180. Using the matrix method, find the number of students allocated to each club.

Formula / idea: Form $AX = B$; if $|A| \neq 0$, then $X = A^{-1}B$ with $A^{-1} = \dfrac{1}{|A|}\operatorname{adj} A$.

1 Let the numbers in Sports, Music and Drama be x, y, z.

2 Write the three conditions as equations. ✓ 1

$x - y - z = 0, \qquad -x + 2y = 40, \qquad x + y + z = 180$

$y = \frac{x}{2} + 20 \Rightarrow 2y = x + 40$

3 Write the system as $AX = B$.

$$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 40 \\ 180 \end{bmatrix}$$

4 Find $|A|$ by expanding along R_1. ✓ 1

$|A| = 1(2 - 0) + 1(-1 - 0) - 1(-1 - 2) = 2 - 1 + 3 = 4 \neq 0$

So A^{-1} exists and the solution is unique.

5 Find the cofactors of row 1. ✓ ½

$A_{11} = 2, \quad A_{12} = 1, \quad A_{13} = -3$

6 Find the cofactors of rows 2 and 3. ✓ ½

$A_{21} = 0,\ A_{22} = 2,\ A_{23} = -2; \qquad A_{31} = 2,\ A_{32} = 1,\ A_{33} = 1$

7 Write $\operatorname{adj} A$ (transpose of cofactors) and A^{-1}. ✓ ½

$$\operatorname{adj} A = \begin{bmatrix} 2 & 0 & 2 \\ 1 & 2 & 1 \\ -3 & -2 & 1 \end{bmatrix}, \qquad A^{-1} = \frac{1}{4}\begin{bmatrix} 2 & 0 & 2 \\ 1 & 2 & 1 \\ -3 & -2 & 1 \end{bmatrix}$$

8 Find $X = A^{-1}B$. ✓ 1

$$X = \frac{1}{4}\begin{bmatrix} 0 + 0 + 360 \\ 0 + 80 + 180 \\ 0 - 80 + 180 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 360 \\ 260 \\ 100 \end{bmatrix} = \begin{bmatrix} 90 \\ 65 \\ 25 \end{bmatrix}$$

9 Hence Sports $= 90$, Music $= 65$, Drama $= 25$ students. ✓ ½

Answer **Sports $= 90$, Music $= 65$, Drama $= 25$ students**

Watch out: Writing condition (ii) as $y = \frac{1}{2}(x + 20)$; it is $y = \frac{x}{2} + 20$.

16 **Long answer** 5 marks

Find the product $\begin{bmatrix}1 & -1 & 2\\ 0 & 2 & -3\\ 3 & -2 & 4\end{bmatrix}\begin{bmatrix}-2 & 0 & 1\\ 9 & 2 & -3\\ 6 & 1 & -2\end{bmatrix}$ and use it to solve: $x - y + 2z = 1$, $2y - 3z = 1$, $3x - 2y + 4z = 2$.

Formula / idea: If $AB = I$, then $A^{-1} = B$.

1 Call the first matrix A and the second B. Multiply row by column. ✓ 1½

$$AB = \begin{bmatrix}-2-9+12 & 0-2+2 & 1+3-4\\ 0+18-18 & 0+4-3 & 0-6+6\\ -6-18+24 & 0-4+4 & 3+6-8\end{bmatrix}$$

2 Simplify the product. ✓ ½

$$AB = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = I$$

3 Since $AB = I$, B is the inverse of A. ✓ 1

$$A^{-1} = B = \begin{bmatrix}-2 & 0 & 1\\ 9 & 2 & -3\\ 6 & 1 & -2\end{bmatrix}$$

4 Write the system as $AX = C$; its coefficient matrix is A. ✓ ½

$$\begin{bmatrix}1 & -1 & 2\\ 0 & 2 & -3\\ 3 & -2 & 4\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}1\\ 1\\ 2\end{bmatrix}$$

5 Find $X = A^{-1}C = BC$. ✓ 1

$$X = \begin{bmatrix}-2 & 0 & 1\\ 9 & 2 & -3\\ 6 & 1 & -2\end{bmatrix}\begin{bmatrix}1\\ 1\\ 2\end{bmatrix} = \begin{bmatrix}-2+0+2\\ 9+2-6\\ 6+1-4\end{bmatrix} = \begin{bmatrix}0\\ 5\\ 3\end{bmatrix}$$

6 Hence $x = 0$, $y = 5$, $z = 3$. ✓ ½

Answer **Product** $= I$; $x = 0$, $y = 5$, $z = 3$

Watch out: Forgetting the 0 coefficient of x in $2y - 3z = 1$ when writing row 2 of A.

17 **Long answer** 5 marks

If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$, find A^{-1}. Hence solve: $x + 2y - 3z = -4$, $2x + 3y + 2z = 2$, $3x - 3y - 4z = 11$.

Formula / idea: $A^{-1} = \dfrac{1}{|A|}\operatorname{adj} A$ and $X = A^{-1}B$.

1 Find $|A|$ by expanding along R_1. ✓ 1

$|A| = 1(-12 + 6) - 2(-8 - 6) - 3(-6 - 9) = -6 + 28 + 45 = 67 \neq 0$

2 Find the cofactors of row 1. ✓ ½

$A_{11} = -6, \quad A_{12} = 14, \quad A_{13} = -15$

3 Find the cofactors of row 2. ✓ ½

$A_{21} = 17, \quad A_{22} = 5, \quad A_{23} = 9$

4 Find the cofactors of row 3. ✓ ½

$A_{31} = 13, \quad A_{32} = -8, \quad A_{33} = -1$

5 Write $\operatorname{adj} A$ (transpose of cofactors) and A^{-1}. ✓ ½

$$\operatorname{adj} A = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}, \qquad A^{-1} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

6 The system is $AX = B$ with this A, so $X = A^{-1}B$. ✓ ½

$$B = \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}, \qquad X = A^{-1}B$$

7 Multiply. ✓ 1

$$X = \frac{1}{67}\begin{bmatrix} 24 + 34 + 143 \\ -56 + 10 - 88 \\ 60 + 18 - 11 \end{bmatrix} = \frac{1}{67}\begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

8 Hence $x = 3$, $y = -2$, $z = 1$. ✓ ½

Answer $A^{-1} = \dfrac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$; $x = 3$, $y = -2$, $z = 1$

Watch out: Using the cofactor matrix itself as $\operatorname{adj} A$; it must be transposed.

18 **Long answer** 5 marks

The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹90. The cost of 6 kg onion, 2 kg wheat and 3 kg rice is ₹70. Find the cost of each item per kg by the matrix method.

Formula / idea: Form $AX = B$; if $|A| \neq 0$, then $X = A^{-1}B$.

1	Let onion, wheat and rice cost ₹x, ₹y, ₹z per kg.	
2	Write the equations. $4x + 3y + 2z = 60, \quad 2x + 4y + 6z = 90, \quad 6x + 2y + 3z = 70$	✓ 1
3	Write the system as $AX = B$. $\begin{bmatrix} 4 & 3 & 2 \\ 2 & 4 & 6 \\ 6 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix}$	
4	Find $\lvert A \rvert$ by expanding along R_1. $\lvert A \rvert = 4(12 - 12) - 3(6 - 36) + 2(4 - 24) = 0 + 90 - 40 = 50 \neq 0$	✓ 1
5	Find the cofactors of row 1. $A_{11} = 0, \quad A_{12} = 30, \quad A_{13} = -20$	✓ ½
6	Find the cofactors of rows 2 and 3. $A_{21} = -5,\ A_{22} = 0,\ A_{23} = 10; \qquad A_{31} = 10,\ A_{32} = -20,\ A_{33} = 10$	✓ ½
7	Write $\operatorname{adj} A$ (transpose of cofactors) and A^{-1}. $\operatorname{adj} A = \begin{bmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{bmatrix}, \qquad A^{-1} = \frac{1}{50} \begin{bmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{bmatrix}$	✓ ½
8	Find $X = A^{-1}B$. $X = \frac{1}{50} \begin{bmatrix} 0 - 450 + 700 \\ 1800 + 0 - 1400 \\ -1200 + 900 + 700 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 250 \\ 400 \\ 400 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix}$	✓ 1
9	Hence onion costs ₹5 per kg, wheat ₹8 per kg and rice ₹8 per kg.	✓ ½

Answer **Onion ₹5 per kg, wheat ₹8 per kg, rice ₹8 per kg**

Watch out: Mixing up the order of items in a row; keep onion, wheat, rice in the same order.

19 **Long answer** 5 marks

If $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, find A^{-1}. Using A^{-1}, solve: $x - 2y = 10$, $2x - y - z = 8$, $-2y + z = 7$.

Formula / idea: The coefficient matrix here is A', and $(A')^{-1} = (A^{-1})'$.

1 Find $|A|$ by expanding along R_1. ✓ 1
$|A| = 1(-1-2) - 2(-2-0) + 0 = -3 + 4 = 1 \neq 0$

2 Find the cofactors of row 1. ✓ ½
$A_{11} = -3, \quad A_{12} = 2, \quad A_{13} = 2$

3 Find the cofactors of row 2. ✓ ½
$A_{21} = -2, \quad A_{22} = 1, \quad A_{23} = 1$

4 Find the cofactors of row 3. ✓ ½
$A_{31} = -4, \quad A_{32} = 2, \quad A_{33} = 3$

5 Write $\text{adj}\, A$ and A^{-1} (here $|A| = 1$). ✓ ½
$A^{-1} = \frac{1}{1}\,\text{adj}\, A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$

6 Write the system in matrix form $CX = B$.
$\begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$

7 The rows of C are the columns of A, so $C = A'$. ✓ ½
$A'X = B \Rightarrow X = (A')^{-1}B = (A^{-1})'B$
Inverse of a transpose is the transpose of the inverse.

8 Multiply. ✓ 1
$X = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix} = \begin{bmatrix} -30 + 16 + 14 \\ -20 + 8 + 7 \\ -40 + 16 + 21 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ -3 \end{bmatrix}$

9 Hence $x = 0$, $y = -5$, $z = -3$. ✓ ½

Answer $A^{-1} = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$; $x = 0$, $y = -5$, $z = -3$

Watch out: Using $X = A^{-1}B$ directly; the coefficient matrix is A', not A.

20 **Case study** 4 marks

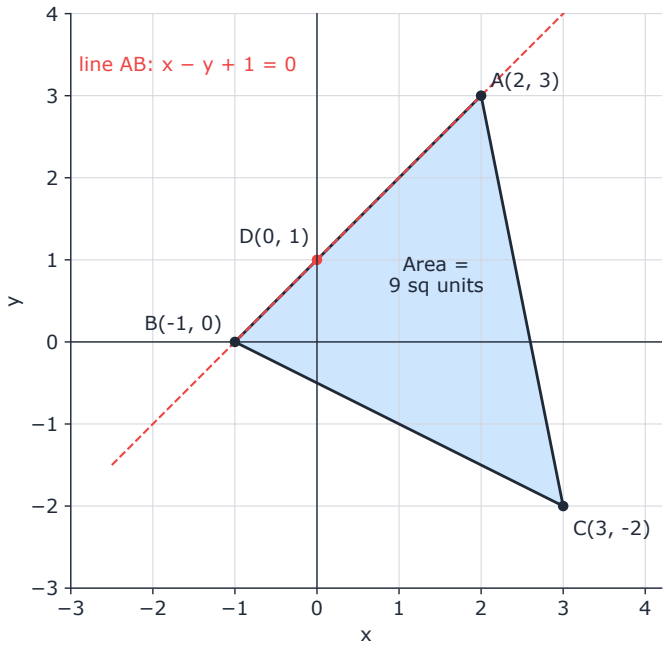
Case Study: A farmer owns a triangular plot of land whose corners, on a map grid, are $A(2,3)$, $B(-1,0)$ and $C(3,-2)$ (1 unit = 10 m). He plans to fix a pole at a point $D(k,1)$ on the boundary line AB.

(i) Using determinants, find the area of the plot in square units. 1 mark

(ii) Find the value of k so that D lies on line AB. 1 mark

(iii) Using determinants, find the equation of line AB. 2 marks

Formula / idea: Area $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$; collinear points give area zero.



(i) 1 mark

1 Write the area as a determinant. ✓ ½

$$\Delta = \frac{1}{2}\begin{vmatrix} 2 & 3 & 1 \\ -1 & 0 & 1 \\ 3 & -2 & 1 \end{vmatrix}$$

2 Expand along R_1.

$$\Delta = \frac{1}{2}\left[2(0+2) - 3(-1-3) + 1(2-0)\right] = \frac{1}{2}(4+12+2) = 9$$

3 Hence area of the plot $= 9$ sq units. ✓ ½

On the ground this is $9 \times 100 = 900$ m².

Answer (i) 9 **sq units**

(ii) 1 mark

1 D on line AB means A, B, D are collinear, so area is zero. ✓ ½

$$\begin{vmatrix} 2 & 3 & 1 \\ -1 & 0 & 1 \\ k & 1 & 1 \end{vmatrix} = 0$$

2 Expand along R_1 and solve.

$$2(0-1) - 3(-1-k) + 1(-1-0) = 0 \;\Rightarrow\; 3k = 0$$

3 Hence $k = 0$, so the pole is at $D(0, 1)$. ✓ ½

Answer (ii) $k = 0$

(iii) 2 marks

1 Let $P(x, y)$ be any point on AB; P, A, B are collinear. ✓ 1

$$\begin{vmatrix} x & y & 1 \\ 2 & 3 & 1 \\ -1 & 0 & 1 \end{vmatrix} = 0$$

2 Expand along R_1. ✓ ½

$$x(3-0) - y(2+1) + 1(0+3) = 0 \;\Rightarrow\; 3x - 3y + 3 = 0$$

3 Divide by 3 to get the equation of line AB. ✓ ½

$$x - y + 1 = 0$$

Answer (iii) $x - y + 1 = 0$

Watch out: In (ii), using C instead of B; D must be collinear with A and B.